

SIZING SHARP WEAR

M1 DIFFERENTIAL OR M2 ABSOLUTE?

Wall-loss direction, very sharp, right at the support edge, and the absolute reads much deeper.
Which number do you use?

KEY IDEA

Pick the code first. The code picks the channel. WAR is sized on M2 absolute, after you check the number isn't inflated.

	Differential (M1)	Absolute (M2)
SEES	Only the change between two close coils	The whole change in one coil
BEST FOR	Small, localized flaws like pits	Gradual or flat wall loss like wear and thinning
SHARP OR LONG FLAWS	Tends to under-call: sees edges, coils partly cancel	Closer to the full depth
WEAK SPOTS	Misses gradual change	Residual, wobble, drift, and where you put the endpoints

1 DECIDE THE CODE

- Wall loss at a support from contact: WAR, sized on M2.
- A localized pit that happens to sit at the support: ODI, called on M1.
- The code decides the channel, not which number looks more believable.

NEVER

- ✗ Average the two channels
- ✗ Pick the smaller number to avoid a call
- ✗ Size wear on a drilled-hole curve without checking procedure

2 SANITY-CHECK M2

- Measure M2 at clean supports in the same tube. A baseline offset there means part of your reading is residual.
- Re-check the null and measurement points. On a sharp signal a small shift changes the percentage a lot.
- Confirm M2 was calibrated per procedure. Many sites use a wear standard, not drilled holes, for the wear curve.

ESCALATE WHEN

- › M2 and M1 disagree by a lot after your checks
- › You can't tell wear from a pit

3 IF IT STILL READS MUCH DEEPER

- Don't quietly pick the smaller number.
- The usual NDE approach is to report conservatively and flag it. Your procedure decides.
- A rotating-probe exam would settle the shape and depth.

QUICK TIPS

Code first, channel second.

Big channel disagreement is a reason to flag on its own.

Your procedure decides how conservative to be.